

ON THE COMBINATORIAL HOPF ALGEBRA OF SIGNED GRAPHS

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- OUTLINE OF THE TALK -

I. PRELIMINARIES ON CHA'S
(COMBINATORIAL HOPF ALGEBRAS)

II. PRELIMINARIES ON SIGNED GRAPHS

III. THE CHA OF SIGNED GRAPHS

IV. DEFORMATIONS OF THE CHARACTER

I. PRELIMINARIES ON COMBINATORIAL HOPF ALGEBRAS

WHAT IS A COMBINATORIAL HOPF ALGEBRA?

Is a pair $(\mathcal{H}, \zeta)$ where:

* $\mathcal{H} = (\mathcal{H}, \mu, \nu, \Delta, \epsilon)$ is a Hopf algebra

In this talk:

- $\mathcal{H}$ vector space over $\mathbb{K}$ with combinatorial objects as bases
- $\mu: \mathcal{H} \otimes \mathcal{H} \longrightarrow \mathcal{H}$
how to merge them?
- $\Delta: \mathcal{H} \longrightarrow \mathcal{H} \otimes \mathcal{H}$
how to split them?
- $\nu: \mathbb{K} \longrightarrow \mathcal{H}, \epsilon: \mathcal{H} \longrightarrow \mathbb{K}$
who is the unit?
- all these maps satisfying a bunch of compatibility axioms

WHAT IS A COMBINATORIAL HOPF ALGEBRA?

Is a pair (H, ζ) where:

* $H = (H, \mu, \nu, \Delta, \epsilon)$ is a Hopf algebra

* H is graded and connected

* $\dim(H_d) < \infty \quad \forall d \geq 0$

* $\zeta: H \longrightarrow \mathbb{K}$ the character is a morphism of $\mathbb{K}$ -algebras

$$\Delta(H_d) \subseteq \sum_{i+j=d} H_i \otimes H_j$$

$$H = \bigoplus_{n \geq 0} H_n, \quad \mu(H_i \otimes H_j) \subseteq H_{i+j}$$

$$\dim(H_0) = 1$$

Example: there exist C.H.A. defined over

$\mathbb{K}[x]$, Sym, QSym, permutations, posets, matroids, graphs...

THE C.H.A. OF GRAPHS

Note $\mathcal{G} := \text{span}_{\mathbb{K}} \left\{ \begin{array}{l} \text{isomorphism} \\ \text{classes of graphs} \end{array} \right\}$

$$\underbrace{3 \cdot (\triangle) - \frac{1}{2} (\bullet \rightarrow \bullet)}_{\in \mathcal{G}_3} - \underbrace{(\bullet)}_{\in \mathcal{G}_1}$$

we can grade it by number of vertices $\mathcal{G} = \bigoplus_{n \geq 0} \mathcal{G}_n$

• $\mu(G \otimes H) := G \sqcup H$ $\mu(\square \otimes I) = \square I$ • $\nu(1) = \emptyset$

• $\Delta(G) := \sum_{S \subseteq V(G)} G|_S \otimes G|\bar{S}$

• $\epsilon(G) = \begin{cases} 1 & \text{if } G = \emptyset \\ 0 & \text{otherwise} \end{cases}$

$$\Delta(\bullet \rightarrow \bullet) = \bullet \rightarrow \bullet \otimes \emptyset + 2(\bullet \otimes \bullet \rightarrow \bullet) + \bullet \otimes \bullet \rightarrow \bullet + 2(\bullet \rightarrow \bullet \otimes \bullet) + \bullet \rightarrow \bullet \otimes \bullet + \emptyset \otimes \bullet \rightarrow \bullet$$

• $\zeta(G) := \begin{cases} 1 & \text{if } G \text{ has no edges} \\ 0 & \text{otherwise} \end{cases}$

* Proposition: $(\mathcal{G}, \mu, \nu, \Delta, \epsilon)$
together with ζ is a C.H.A.

ALGEBRAIC INVARIANTS OF COMB. OBJECTS

* THEOREM [Aguilar, Bergeron & Sottile '03]

For any C.H.A. $(\mathcal{H}, \zeta)$, there exists a unique morphism of C.H.A.

$$\Psi: (\mathcal{H}, \zeta) \longrightarrow (\text{QSym}, \zeta_Q)$$

Example: In $(\mathcal{G}, \zeta)$, for any graph G , $\Psi(G) = \sum_{K \text{ proper}} x_{K(v_1)} \cdots x_{K(v_d)}$

* Remark

$$(\mathcal{H}, \zeta) \xrightarrow{\Psi} (\text{QSym}, \zeta_Q) \xrightarrow[\substack{\text{evaluate in} \\ (1, \dots, 1, 0, \dots)}]{\substack{\phi_t := \\ t}} (\mathbb{K}[t], \zeta_P)$$

Stanley's chromatic
symmetric function

Example: In $(\mathcal{G}, \zeta)$, for any graph G :

$$\phi_t \circ \Psi(G) = \# \left\{ \begin{array}{l} \text{proper colorings of } G \\ \text{using colors } \{1, \dots, t\} \end{array} \right\} = \chi_G(t)$$

ALGEBRAIC INVARIANTS OF COMB. OBJECTS

Recall a *character* in a graded connected Hopf algebra $\mathcal{H}$ is a morphism of algebras $\zeta: \mathcal{H} \longrightarrow \mathbb{K}$

- Def: The *convolution product* $*$ of two characters ζ_1, ζ_2 is the character defined by

$$\zeta_1 * \zeta_2 = m \circ (\zeta_1 \otimes \zeta_2) \circ \Delta$$

- Proposition: The composite map $(\mathcal{H}, \zeta) \xrightarrow{\Psi} (\mathcal{QSym}, \zeta_{\mathcal{Q}}) \xrightarrow{\phi_t} (\mathbb{K}[t], \zeta_P)$

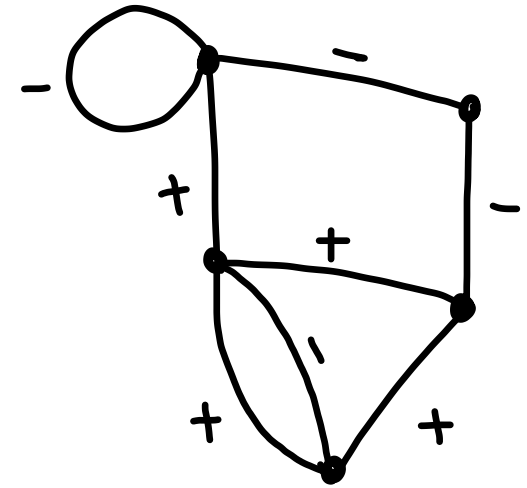
$$\text{is } \underbrace{\zeta * \dots * \zeta}_{t \text{ times}} = \zeta^{*t}$$

in the preceding example:

$$\zeta^{*t}(G) = \chi_G(t)$$

II. PRELIMINARIES

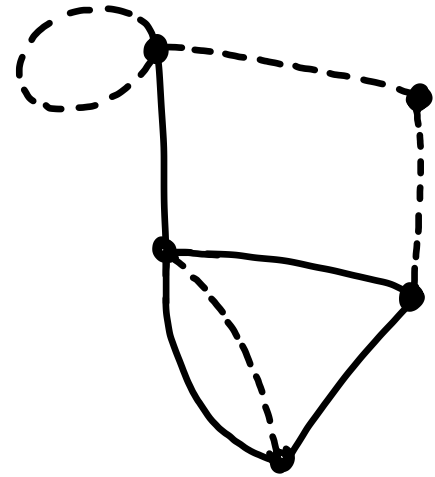
ON SIGNED GRAPHS



II. PRELIMINARIES ON SIGNED GRAPHS

A **signed graph** is a graph $G=(V,E)$ together with a sign function $\sigma: E \rightarrow \{+,-\}$

A signed graph is **balanced**, if every cycle has an even number of negative edges

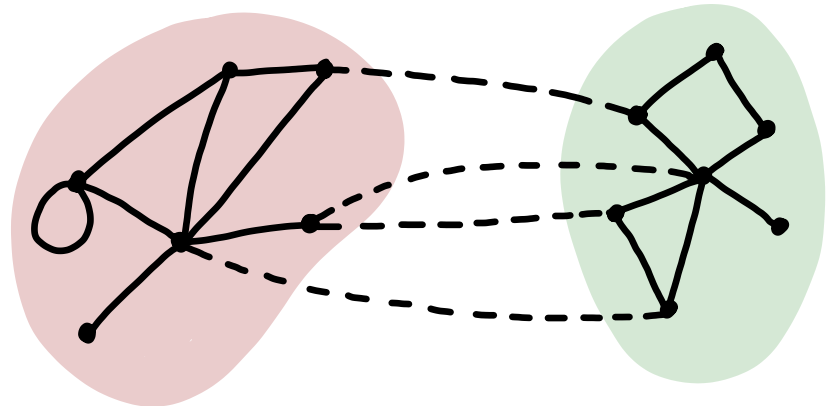


* THEOREM [Harary '54]

A signed graph $\Sigma=(G,\sigma)$ is balanced iff it admits a decomposition

$V = A \cup B$ s.t.

- edges in A or B are positive
- edges between A and B are negative



II. PRELIMINARIES ON SIGNED GRAPHS

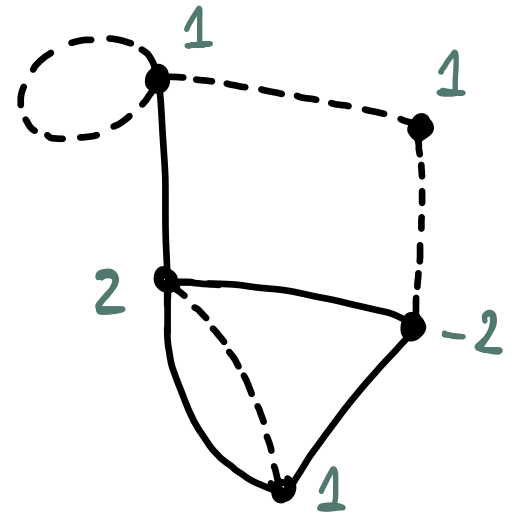
SIGNED GRAPH COLORING

Given a signed graph $\Sigma = (G, \sigma)$ a k -proper coloring of Σ is a map $\kappa: V \rightarrow \{\pm 1, \dots, \pm k\}$ such that for any edge $e = \{i, j\}$, $\kappa(i) \neq \sigma(e) \kappa(j)$

* Definition (Zaslavsky)

Given a signed graph Σ , the signed chromatic polynomial is

$$\chi_{\Sigma}(2k) := \# \left\{ \begin{array}{l} \text{proper } k\text{-colorings} \\ \text{of } \Sigma \end{array} \right\}$$



III. OUR RESEARCH:

THE C.H.A. OF

SIGNED GRAPHS

THE C.H.A. OF SIGNED GRAPHS

$$\mathcal{S} := \text{span}_{\mathbb{K}} \left\{ \begin{array}{l} \text{isomorph classes} \\ \text{of signed graphs} \end{array} \right\}$$

$$s = \triangle + 3(\text{loop}) - (\bullet)$$

The Hopf algebra structure is the same :

$$\begin{aligned} \bullet \mu(\Sigma_1 \otimes \Sigma_2) &:= \Sigma_1 \sqcup \Sigma_2, & \bullet \nu(1) &:= \emptyset \\ \bullet \Delta(\Sigma) &:= \sum_{S \subseteq V} \Sigma|_S \otimes \Sigma|_{S^c} & \bullet \epsilon(\Sigma) &:= \begin{cases} 1 & \text{if } \Sigma = \emptyset \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

notation

$$\Sigma = (G, \sigma)$$

$$\bar{\Sigma} = (G, -\sigma)$$

And for connected signed graphs :

$$\zeta(\Sigma) := \begin{cases} 2 & \text{if } \bar{\Sigma} \text{ balanced} \\ 0 & \text{otherwise} \end{cases}$$

Define $(\mathcal{S}, \zeta)$ the CHA of signed graphs

THE CANONICAL MORPHISM

* THEOREM: Let Ψ be the only morphism of CHA from (S, ζ) to $(QSym, \zeta_Q)$, then, for any signed graph $\Sigma = ([d], E, \sigma)$

$$\Psi(\Sigma) = \sum_{\kappa \text{ prop.}} \chi_{|\kappa(1)|} \cdots \chi_{|\kappa(d)|}$$

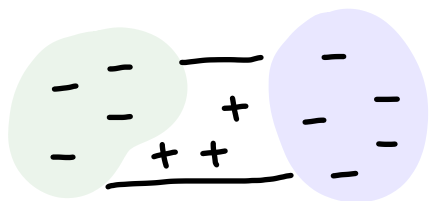
Recall:

$$(H, \zeta) \xrightarrow{\Psi} (QSym, \zeta_Q) \xrightarrow[\substack{\text{eval} \\ (1, \dots, 1, 0 \dots)}]{} (K[t], \zeta_P)$$


* Corollary: $\zeta^{*t}(\Sigma) = \chi_{\Sigma}(2t) = \# \left\{ \begin{array}{l} \kappa: V \rightarrow \{\pm 1, \dots, \pm t\} \\ \text{proper} \end{array} \right\}$

SKETCH OF THE PROOF

*Lemma: For any signed graph $\Sigma = (V, E, \sigma)$, $\zeta(\Sigma)$ counts the number of decompositions $V = A \sqcup B$ in which:



- edges between blocks are +
- edges on the same block are -

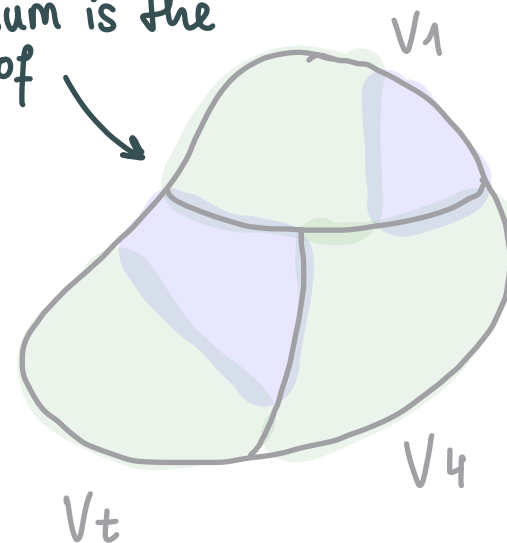
Proof (of the corollary)

$$\zeta^{*t}(\Sigma) = m^{(t-1)} \cdot \zeta^{\otimes t} \cdot \Delta^{(t-1)}(\Sigma)$$

theorem $\uparrow$
ABS

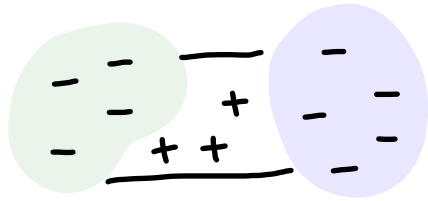
$$= \sum_{V = V_1 \sqcup \dots \sqcup V_t} \zeta(G|_{V_1}) \dots \zeta(G|_{V_t})$$

this sum is the number of



SKETCH OF THE PROOF

*Lemma: For any signed graph $\Sigma = (V, E, \sigma)$, $\zeta(\Sigma)$ counts the number of decompositions $V = A \sqcup B$ in which:



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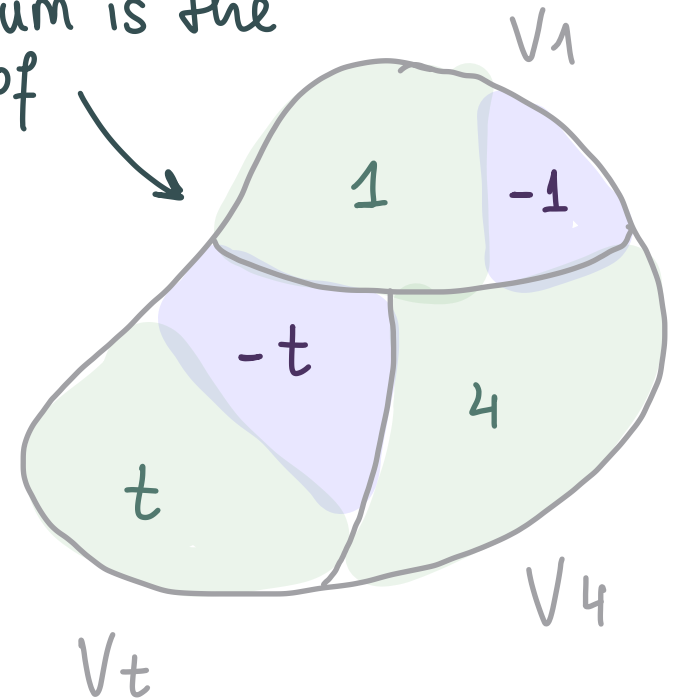
Proof (of the corollary)

$$\zeta^{*t}(\Sigma) = \sum_{V = V_1 \sqcup \dots \sqcup V_t} \zeta(\Sigma|_{V_1}) \dots \zeta(\Sigma|_{V_t})$$

theorem
ABS

But these are in
bijection with
proper t -colorings

this sum is the
number of



RECIPROCITY RESULTS VIA THE ANTIPODE

$$\begin{array}{ccc}
 (S, \zeta) & \xrightarrow{\zeta^{*t}} & (\mathbb{K}[t], \zeta_P) \\
 S \downarrow & \curvearrowright & \downarrow S \\
 (S, \zeta) & \xrightarrow{\zeta^{*t}} & (\mathbb{K}[t], \zeta_P)
 \end{array}$$

$$S \circ \zeta^{*t}(\Sigma) = \zeta^{*t} \circ S(\Sigma)$$

$\Downarrow$

$$S \circ \chi_{\Sigma}(2t) = \zeta^{*t} \circ S(\Sigma)$$

$\Downarrow$

$$\chi_{\Sigma}(-2t) = \zeta^{*t} \circ S(\Sigma)$$

Takeuchi's formula

* THEOREM [Zaslavsky]

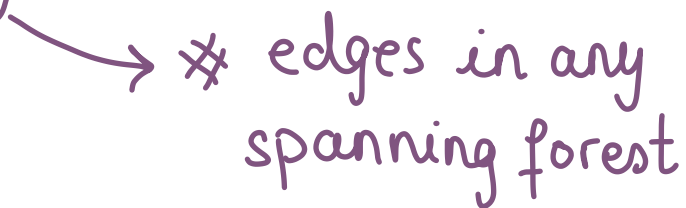
$$\chi_{\Sigma}(-2t) = \sum \left\{ (K, \square), \square \text{ acyclic orientation} \right. \\ \left. K: V \rightarrow \{\pm 1, \dots, \pm t\} \text{ compatible} \right\}$$

IV. DEFORMATIONS OF THE CHARACTER

DEFORMATION OF A CHARACTER

- Def: Given a character $\zeta: \mathcal{H} \longrightarrow \mathbb{K}$, a deformation of ζ is a morphism of algebras $\zeta_x: \mathcal{H} \longrightarrow \mathbb{K}[x]$ such that $\zeta_x(h)(0) = \zeta(h) \quad \forall h \in \mathcal{H}$

Ex [BHM16]: In the CHA of graphs $\zeta(G) = \begin{cases} 1 & \text{if } G \text{ has no edges} \\ 0 & \text{otherwise} \end{cases}$

define $\zeta_x(G) := x^{\text{rk}(G)}$  $\ast$ edges in any spanning forest

multiplicative as $\text{rk}(G_1 \sqcup G_2) = \text{rk}(G_1) + \text{rk}(G_2)$ and clearly

$$\zeta(G) = 0^{\text{rk}(G)}$$

IV. DEFORMATIONS OF THE CHARACTER

THE WHITNEY NUMBER POLYNOMIAL

- Definition (Zaslavsky). Given a graph $G=(V,E)$ the Whitney number polynomial is

$$w_G(x,t) = \sum_{A \in \mathcal{F}(G)} x^{\text{rk}(A)} \cdot \chi_{G/A}(t)$$

$\mathcal{F}(G) \rightarrow \text{flats of } G$

It has interesting specializations, for instance

$$(-1)^{|V|} w_G(-x, -1) = \# \text{ polynomial of } \mathbb{Z}[G]$$

- Proposition: $\sum_x^{*t} (G) = w_G(x,t)$

obs: $w_G(0,t) = \chi_G(t)$

Recall the convolution of two characters is

$$\zeta_1 * \zeta_2 := m \circ (\zeta_1 \otimes \zeta_2) \circ \Delta$$

THE DICHROMATIC POLYNOMIAL

The **chromatic polynomial** is the polynomial generated by:

$$\chi_G(t) = \chi_{G \setminus e}(t) - \chi_{G/e}(t) \quad D_G(u, t) = D_{G \setminus e}(u, t) + u D_{G/e}(u, t)$$

$$\chi_{G \sqcup H}(t) = \chi_G(t) \cdot \chi_H(t)$$

$$D_{G \sqcup H}(u, t) = D_G(u, t) \cdot D_H(u, t)$$

$$\chi_G(t) = t \quad \text{if } G = \bullet$$

$$D_G(u, t) = t \quad \text{if } G = \bullet$$

The **dichromatic polynomial** is the polynomial $D_G(x, t)$ generated by

- Rmk 1: $\chi_G(t) = D_G(-1, t)$

- Rmk 2: The dichromatic polynomial is related to the **Tutte polynomial** as:

$$D_G(u, t) = t^{cc(G)} u^{|V| - cc(G)} T_G(1 + ut^{-1}, 1 + t) \quad [K89]$$

THE DICHROMATIC POLYNOMIAL

* Proposition: Let ζ_x be the deformation of ζ defined by

$\zeta_x(G) := x^{|E|}$ for any graph $G = (V, E)$, then

$$\zeta_x^{*t}(G) = D(x-1, t)$$

IDEA OF THE PROOF:

First show that $D_G(x, t) = \sum (1+x)$

$|I(K)|$  number of monochromatic edges of K

$$\zeta_x^{*t}(G) = \sum_{V_1 \sqcup \dots \sqcup V_t} \zeta_x(G|_{V_1}) \cdots \zeta_x(G|_{V_t}) = \sum_{V_1 \sqcup \dots \sqcup V_t} x^{|E(G|_{V_1})| + \dots + |E(G|_{V_t})|}$$

AN EXAMPLE

Let $G = \text{---} \bigcirc \text{---} \bigcirc \text{---}$, the convolution $\zeta_x^{*t}(G)$ can be computed

$$\text{as: } \zeta_x^{*t}(G) = \sum_{\alpha \models 3} \binom{t}{\ell(\alpha)} \sum_{\substack{V = V_1 \sqcup \dots \sqcup V_{\ell(\alpha)} \\ |V_i| = \alpha_i}} \zeta_x(G|_{V_1}) \cdots \zeta_x(G|_{V_{\ell(\alpha)}})$$

• $\alpha = 3 \Rightarrow \binom{t}{1} \cdot \zeta_x(G) = t \cdot x^4$

• $\alpha = 2, 1 \Rightarrow 2 \cdot \binom{t}{2} (\zeta_x(\text{---} \bigcirc \text{---} \bigcirc) + \zeta_x(\text{---} \bigcirc \text{---} \bigcirc) + \zeta_x(\bigcirc \text{---} \bigcirc)) =$
 $= t(t-1)(x^2 + x + x^3)$

• $\alpha = 1, 1, 1 \Rightarrow \binom{t}{3} \cdot 3! \cdot \zeta_x(\bigcirc \text{---} \bigcirc \text{---} \bigcirc) = t(t-1)(t-2) \cdot x$

$$T_G(y, s) = s^2 y + y s^2$$

And after the suitable change of variables $\nearrow$

SOME BIBLIOGRAPHY

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Merci

beaucoup!



DES QUESTIONS?